

Demonstration of Proposed Features

Phase 8 · Project Demonstration

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This document is the ordered walkthrough script for demonstrating the Script-Controlled ACL project. It proves that record access to the `u_institution_details` table depends on both the user's role and the Branch field value, using ServiceNow impersonation as the evidence. Each step names the feature shown and exactly what the audience sees on screen.

Feature Walkthrough

Step	Feature demonstrated	What the audience sees
1	Elevate to security_admin	Admin uses the elevate-role menu and confirms security_admin, which is required before editing ACLs.
2	Custom table and schema	The <code>u_institution_details</code> table with its columns: Student Roll Number, Student Name (ref), Faculty Name (ref), Branch, Email, Phone Number, Description.
3	Branch choice field	The Branch field offering exactly three choices: ECE, EEE, CSE.
4	Sample data	Four records in the list: ECE001 (ECE), EEE001 (EEE), EEE002 (EEE), CSE001 (CSE).
5	ACL inventory	Four record-level ACLs on the table, one each for read, create, write, delete.
6	READ ACL detail	The read ACL set to Advanced, requiring role bb1, with data condition Branch is EEE, and the <code>gs.hasRole</code> script.
7	Create/Write/Delete ACLs	Create requires bb2, write requires bb3, delete requires bb4, each with no condition and no script.
8	Default role removed	The auto-generated <code>u_institution_details_user</code> role removed so each ACL needs only its bb role.
9	Test user roles	EEE User holds all four roles bb1, bb2, bb3, bb4.
10	Impersonate EEE User (read)	The Institution Details list shows only EEE001 and EEE002, that is 2 of 2 EEE records; the New button is visible.

Step	Feature demonstrated	What the audience sees
11	Impersonate EEE User (manage)	Opening EEE001 shows an editable form with both Update and Delete available.
12	Impersonate Abel Tuter (no roles)	The list is replaced by the message: Security constraints prevent access to requested page.
13	Return as System Administrator	As admin the list shows all four records, confirming the script's admin bypass.
14	Wrap-up	Summary that access = role plus Branch value, enforced at record level with no external app.

READ ACL Logic Shown at Step 6

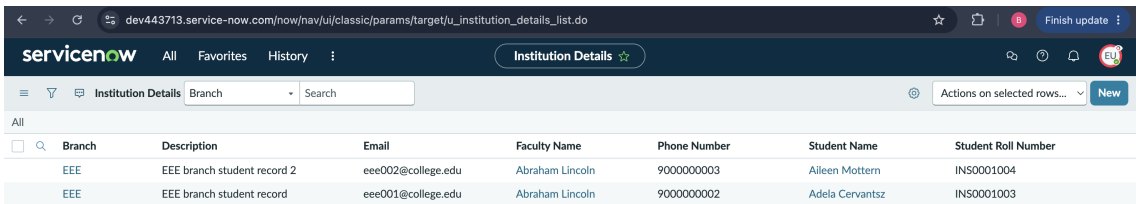
- Requires role bb1 and is marked Advanced.
- Data condition on the ACL: Branch is EEE (u_branch=EEE).
- Script returns true for admin, true for bb1, and false for everyone else:

```
(function(){ if (gs.hasRole('admin')) { return true; } if (gs.hasRole('bb1')) { return true; } return false; })()
```
- Net effect: a bb1 holder reads only rows where Branch is EEE, while admin reads all rows.

Expected Outcomes to Call Out

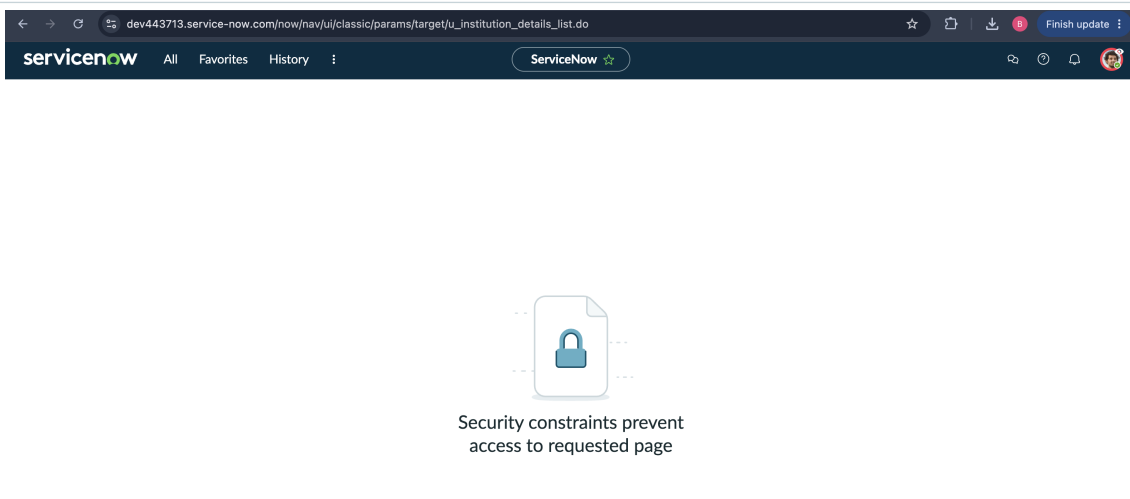
Persona	Roles	Result
EEE User	bb1-bb4	Sees 2 EEE records, can create, update, delete
Abel Tuter	none	Blocked by security constraints
System Administrator	admin	Sees all 4 records

Evidence (from the ServiceNow instance)



Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	EEE branch student record 2	eee002@college.edu	Abraham Lincoln	9000000003	Aileen Mottern	INS0001004
EEE	EEE branch student record	eee001@college.edu	Abraham Lincoln	9000000002	Adela Cervantsz	INS0001003

EEE User sees only the two EEE records



No-role user blocked

A screenshot of a web browser displaying a ServiceNow page. The address bar shows the URL: dev443713.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do. The page header includes the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, and an Institution Details button. Below the header, there is a search bar and a table of institution details. The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. There are four rows of data.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	EEE branch student record 2	eee002@college.edu	Abraham Lincoln	9000000003	Aileen Mottern	INS0001004
EEE	EEE branch student record	eee001@college.edu	Abraham Lincoln	9000000002	Adela Cervantsz	INS0001003
CSE	CSE branch student record	cse001@college.edu	Abraham Lincoln	9000000004	Alejandra Prenatt	INS0001005
ECE	ECE branch student record	ece001@college.edu	Abraham Lincoln	9000000001	Abel Tuter	INS0001002

Admin sees all four records